1 I

1.1 (1)

The time-independent Schrödinger equation (TISE) is

$$-\frac{\hbar^2}{2m}\nabla^2\psi + V(r)\psi = E_1\psi,$$

where the Coulomb potential is

$$V(r) = -\frac{e^2}{4\pi\varepsilon_0 r}.$$

In spherical coordinates, for a wave function depending only on r, the radial part of the Laplacian is

$$\begin{aligned}\nabla^2\psi &= \frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{d\psi}{dr}\right)\\ &= \frac{2}{r}\frac{d\psi}{dr} + \frac{d^2\psi}{dr^2}.\end{aligned}$$

Specifically, substituting the given wave function $\psi_1(r) = e^{-r/a}$ into the TISE gives

$$-\frac{\hbar^2}{2m}\left[\frac{2}{r}\frac{d}{dr} + \frac{d^2}{dr^2}\right]e^{-r/a} - \frac{e^2}{4\pi\varepsilon_0 r}e^{-r/a} = E_1 e^{-r/a}.$$

Noticing that

$$\frac{d}{dr}e^{-r/a} = -\frac{1}{a}e^{-r/a}, \qquad \frac{d^2}{dr^2}e^{-r/a} = \frac{1}{a^2}e^{-r/a},$$

we obtain

$$-\frac{\hbar^2}{2m}\left(-\frac{2}{ar} + \frac{1}{a^2}\right) - \frac{e^2}{4\pi\varepsilon_0 r} = E_1.$$

Rearranging,

$$\left[\frac{\hbar^2}{ma} - \frac{e^2}{4\pi\varepsilon_0}\right]\frac{1}{r} - \frac{\hbar^2}{2ma^2} = E_1.$$

Since E_1 must be independent of r, the coefficient of $1/r$ must vanish:

$$\frac{\hbar^2}{ma} - \frac{e^2}{4\pi\varepsilon_0} = 0.$$

Thus,

$$\boxed{a = \frac{4\pi\varepsilon_0\hbar^2}{me^2}}.$$

Substituting this into the remaining term gives

$$\boxed{E_1 = -\frac{\hbar^2}{2ma^2} = -\frac{me^4}{32\pi^2\varepsilon_0^2\hbar^2}}.$$

2 II

2.1 (2)

Using the Heisenberg equation of motion,

$$\frac{d\vec{L}}{dt} = \frac{1}{i\hbar}[\vec{L}, H],$$

where the orbital angular momentum operator has no explicit time dependence.

For a general central potential,

$$H = \frac{\vec{p}^2}{2m} + V(r), \qquad \vec{L} = \vec{r} \times \vec{p}.$$

We can expand the commutator as

$$\begin{aligned}[\vec{L}, H] &= \left[\vec{r} \times \vec{p}, \frac{\vec{p}^2}{2m} + V(r)\right] \\ &= [\vec{r}, H] \times \vec{p} + \vec{r} \times [\vec{p}, H].\end{aligned}$$

Using the canonical commutation relations,

$$[\vec{r}, H] = \frac{i\hbar}{m}\vec{p}, \qquad [\vec{p}, H] = -i\hbar\nabla V(r),$$

we obtain

$$\begin{aligned}[\vec{L}, H] &= \frac{i\hbar}{m}\vec{p} \times \vec{p} - i\hbar\vec{r} \times \nabla V(r) \\ &= 0.\end{aligned}$$

Here, $\vec{p} \times \vec{p} = 0$, and $\nabla V(r)$ is parallel to $\vec{r}$ because the potential is central.

Therefore,

$$\boxed{\frac{d\vec{L}}{dt} = 0}.$$

Thus, the orbital angular momentum is conserved.

2.2 (3)

The z component of the orbital angular momentum is

$$L_z = xp_y - yp_x.$$

Using the position representation $\vec{p} = -i\hbar\nabla$, we have

$$L_z = -i\hbar\left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}\right).$$

In spherical coordinates,

$$\begin{cases} x = r\sin\theta\cos\phi, \\ y = r\sin\theta\sin\phi, \\ z = r\cos\theta. \end{cases}$$

Applying the chain rule,

$$\begin{aligned}\frac{\partial}{\partial\phi} &= \frac{\partial x}{\partial\phi}\frac{\partial}{\partial x} + \frac{\partial y}{\partial\phi}\frac{\partial}{\partial y} + \frac{\partial z}{\partial\phi}\frac{\partial}{\partial z} \\ &= -r\sin\theta\sin\phi\frac{\partial}{\partial x} + r\sin\theta\cos\phi\frac{\partial}{\partial y} \\ &= -y\frac{\partial}{\partial x} + x\frac{\partial}{\partial y}.\end{aligned}$$

Therefore,

$$\boxed{L_z = -i\hbar\frac{\partial}{\partial\phi}}.$$

2.3 (4)

Consider the eigenvalue equation

$$L_z\Phi(\phi) = m_z\Phi(\phi).$$

Using the result from (3),

$$-i\hbar\frac{d\Phi}{d\phi} = m_z\Phi.$$

Solving this differential equation gives

$$\Phi(\phi) = Ce^{im_z\phi/\hbar}.$$

Since the wave function must be single-valued,

$$\Phi(\phi + 2\pi) = \Phi(\phi).$$

Therefore,

$$e^{i2\pi m_z/\hbar} = 1,$$

which requires

$$\frac{2\pi m_z}{\hbar} = 2\pi n, \qquad n \in \mathbb{Z}.$$

Thus, the possible eigenvalues are

$$\boxed{m_z = n\hbar, \qquad n \in \mathbb{Z}}.$$

For a state with a given orbital quantum number ℓ, the allowed values are further restricted to $n = -\ell, -\ell+1, \ldots, \ell$.

3 III

3.1 (5)

The total angular momentum is

$$\vec{J} = \vec{L} + \vec{S}.$$

Since the orbital and spin angular momentum operators commute with each other,

$$\begin{aligned} J^2 &= (\vec{L} + \vec{S})^2 \\ &= L^2 + 2\vec{L} \cdot \vec{S} + S^2. \end{aligned}$$

Thus,

$$\vec{L} \cdot \vec{S} = \frac{1}{2}(J^2 - L^2 - S^2).$$

For an electron, $s = 1/2$, so

$$S^2 = s(s+1)\hbar^2 = \frac{3}{4}\hbar^2.$$

Therefore,

$$\boxed{\vec{L} \cdot \vec{S} = \frac{1}{2}\left(J^2 - L^2 - \frac{3}{4}\hbar^2\right)}.$$

3.2 (6)

For an electron with orbital quantum number $\ell \geq 1$ and spin quantum number $s = 1/2$, the possible total angular momentum quantum numbers are

$$j_+ = \ell + \frac{1}{2}, \qquad j_- = \ell - \frac{1}{2}.$$

The degeneracy of each level is given by $2j + 1$.

For j_+,

$$\boxed{d_+ = 2j_+ + 1 = 2\ell + 2}.$$

For j_-,

$$\boxed{d_- = 2j_- + 1 = 2\ell}.$$

Notice that

$$d_+ + d_- = 2(2\ell + 1),$$

which agrees with the original degeneracy.

3.3 (7)

The spin–orbit interaction is

$$H_{LS} = \lambda g(r) \vec{L} \cdot \vec{S}.$$

Using first-order perturbation theory, the energy correction for a state with total angular momentum quantum number j is

$$E_j^{(1)} = \lambda \langle g(r) \rangle \langle \vec{L} \cdot \vec{S} \rangle.$$

From (5), we have

$$\langle \vec{L} \cdot \vec{S} \rangle = \frac{\hbar^2}{2}\left[j(j+1) - \ell(\ell+1) - \frac{3}{4}\right].$$

Thus,

$$E_j^{(1)} = \frac{\lambda\hbar^2}{2}\langle g(r)\rangle \left[j(j+1) - \ell(\ell+1) - \frac{3}{4} \right].$$

For $j_+ = \ell + 1/2$,

$$\begin{aligned} E_+^{(1)} &= \frac{\lambda\hbar^2}{2}\langle g(r)\rangle \left[\left(\ell + \frac{1}{2}\right)\left(\ell + \frac{3}{2}\right) - \ell(\ell+1) - \frac{3}{4} \right] \\ &= \frac{\lambda\hbar^2}{2}\langle g(r)\rangle\, \ell. \end{aligned}$$

For $j_- = \ell - 1/2$,

$$\begin{aligned} E_-^{(1)} &= \frac{\lambda\hbar^2}{2}\langle g(r)\rangle \left[\left(\ell - \frac{1}{2}\right)\left(\ell + \frac{1}{2}\right) - \ell(\ell+1) - \frac{3}{4} \right] \\ &= -\frac{\lambda\hbar^2}{2}\langle g(r)\rangle(\ell+1). \end{aligned}$$

Therefore, the signed energy difference between the two levels is

$$\begin{aligned} E_+^{(1)} - E_-^{(1)} &= \frac{\lambda\hbar^2}{2}\langle g(r)\rangle \left[\ell + (\ell+1)\right] \\ &= \boxed{\frac{\lambda\hbar^2}{2}(2\ell+1)\langle g(r)\rangle}. \end{aligned}$$

The non-negative level splitting is the absolute value of this expression.